GENERAL GUIDELINES:

a. Paper
   - Size: A4
   - Orientation: Portrait (except for special diagrams)

b. Indentation: 0.5 inch
c. Margins:
   - Top and Left: 1 inch
   - Bottom and Right: 1 inch
   - Header and Footer: 0.5

d. Alignment: Paragraphs are Justified
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   - Sizes
     - Heading 1, 2, 3: 12
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   - Top Right (defined in the template)
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KOLEHIYO NG LUNGSOD NG DASMARIÑAS

ILearn: Development of an Interactive E-Learning Management System with Automated Progress Tracking for CSS Students of Paliparan III Senior High School

AN UNDERGRADUATE CAPSTONE PROPOSAL

Presented to the Faculty of the
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Bachelor of Science in Information Systems

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KOLEHIYO NG LUNGSOD NG DASMARIÑAS

PROJECT OVERVIEW

Background of the Study

In today's digital age, education continues to evolve through the integration of technology in the teaching and learning process. In Paliparan III, the senior high schools particularly within the Computer Systems Servicing (CSS) strand, students rely heavily on accessible learning materials, interactive content, and timely feedback to support both theoretical understanding and practical skills. Nevertheless, there are still many schools that struggle with the distribution of lesson manually, limited student participation and no real-time monitoring of academic progress. These gaps complicate the teachers' class requirement management and the students' keeping track of their learning performance.

According to Duch, May, & George (2024) the use of Learning Management Systems (LMS) has grown in popularity primarily for their ability to manage and organize digital information related to teaching and learning. And to address the growing need for a more efficient and accessible learning environment. This project proposes the development of an Interactive E-Learning Management System for CSS student of Paliparan III Senior High School. The system aims to provide an interactive online platform that enhances both teaching and learning experiences by offering a space for lessons, quizzes and students monitoring. The proposed system allows teachers to upload learning materials, administer assessments, and monitor student participation seamlessly. And one of its key features is an automated progress tracking tool which evaluates student performance based on quiz results and engagement levels. This tool will generate visual reports and dashboards for both teachers and students promoting transparency and a deeper understanding of academic progress. By combining these core functions, the E-Learning Management System seeks to

improve academic monitoring, streamline communication between teachers and learners and foster a more interactive and data-driven educational environment. This project supports the school's commitment to embracing innovation and enhancing the quality of education through technology.

Statement of the Problem

The purpose of this study is to develop an Interactive E-Learning Management System designed to support CSS (Computer Systems Servicing) students of Paliparan III Senior High School in improving learning efficiency, engagement, and performance monitoring. Specifically, it seeks to address the following problems:

1. How can the system provide an organized and accessible digital platform where CSS students can easily access learning materials, modules, and assessments to improve their learning experience?
2. How can the system provide interactive learning tools (such as quizzes and guided tasks) that help students understand CSS concepts more effectively?
3. What solution can be developed to automate the tracking of student progress, allowing teachers to efficiently monitor completion of modules, quiz results, and overall performance in real time?
4. How can teachers be given a real-time and organized view of each student's progress to support timely feedback and intervention?
5. How can the system generate automated progress reports that help both students and teachers identify strengths, weaknesses, and areas needing improvement?

Objectives of the Study

The primary objectives of this study are to develop an E-Learning Management System that provides an efficient online platform for Paliparan III Senior Highschool

students and teachers to manage lessons, quizzes, attendance and performance monitoring through automated progress tracking.

Specific Objectives:

This study aims to:

1. Design and develop an E-Learning Management System for Paliparan III Senior High School that will be capable of:
    a. A user-friendly web-based interface for both teachers and students to easily use the system.
    b. User Authentication and Role Management to ensures secure access and proper identification of users before entering the platform.
    c. Course and Subject Management to display lists of subjects and participants for organized academic structure.
    d. Teacher Dashboard to provide teachers with tools for uploading lessons, managing quizzes and grades, posting announcements, and monitoring student performance efficiently.
    e. Student Dashboard provides students with all learning resources and academic progress in one place. This also allow students to view enrolled subjects, access learning materials, and take quizzes or submit assignments.
    f. Assessment and Evaluation provides teachers to create quizzes, assignments, and grading, enabling automatic scoring and consistent evaluation of student performance.
    g. Automated Progress Tracking – tracks student performance based on quiz results, attendance, and submissions, providing visual progress reports and analytics.

h. Announcements and Notifications to keep both teachers and students informed about important updates, deadlines, and activities through timely notifications and announcements.

i. Calendar and Scheduling to help users stay organized by displaying class schedules, deadlines, and school events, along with reminders for upcoming activities.

j. Communication and Feedback System to promote interaction between teachers and students through direct messaging, comments, and feedback channels for better learning support.

k. Reports and Analytics to generate detailed reports on student progress, teacher performance, and system activity, providing insights for academic improvement and decision-making.

l. Admin Management Tools this allow administrators to have a full control over the system's users, courses, and data, including maintenance tasks like backups, settings, and report generation.

m. Security and Data Integrity to protect user information and system data through encryption, role-based access, and audit logs, ensuring reliability and accountability.

2. Identify the software and hardware requirements necessary for the development and deployment of the system.

3. Test the performance of the system through unit, integration, and usability testing involving:

   a. Teachers

   b. IT professionals

   c. Non-technical users

4. Evaluate the system's quality based on the ISO 25010 Software Quality Standards, focusing on functionality, reliability, usability, and efficiency.

METHODOLOGY AND OUTCOMES

Conceptual Framework

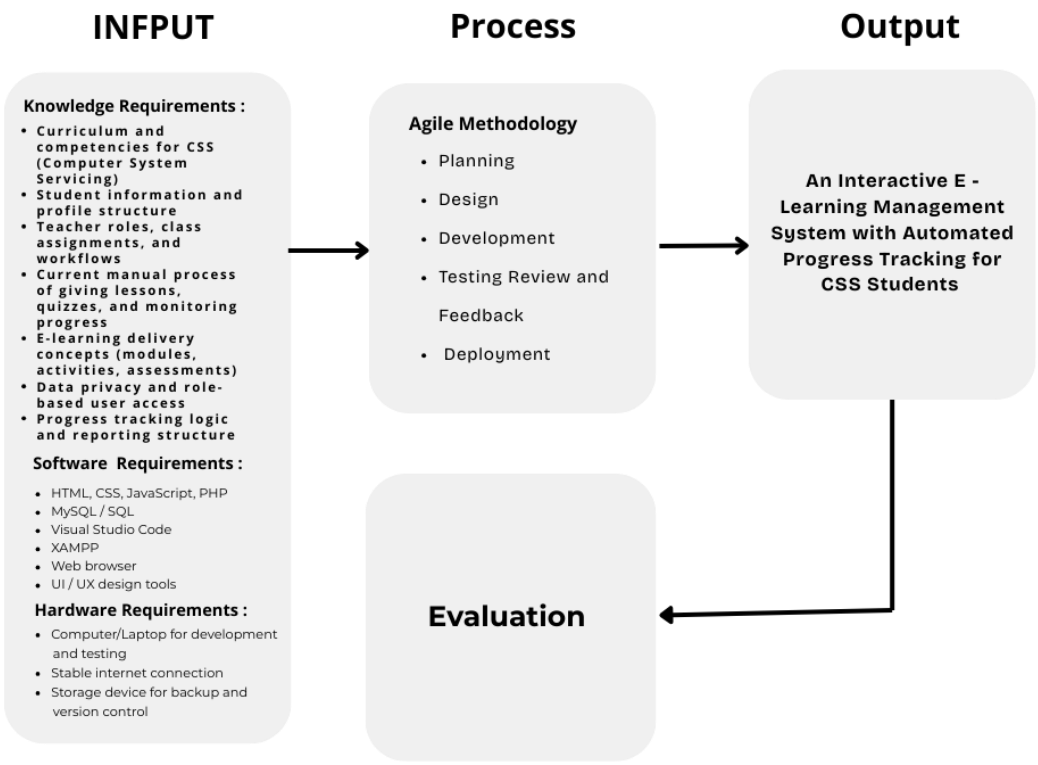

Figure 1

Conceptual framework of Interactive E-Learning Management System with Automated Progress Tracking for CSS Students

Figure 1: The Input phase for ILearn is based on three essential resource categories: Knowledge Requirements, Software Resources, and Hardware Resources.

These combined inputs provide the foundation for the system's architecture and development. It begins with Knowledge Requirements. This phase is where the indispensable data and operational specifications: this means acquiring the complete official CSS curriculum and competencies to make sure the system's content is accurate. It also means writing down the current school practices, like the existing manual method of teaching lessons and recording progress, which the new system will either take over or assimilate. While, the progress tracking logic and data privacy/user access rules are the key factors to making the system workable and safe at the very beginning. The Software Requirements make it clear what the development stack is, for example, PHP for server-side programming, MySQL/SQL for the powerful database that will store the student and curriculum data, and client-side technologies like HTML, CSS, and JavaScript for the interactive part of the website. The Hardware Requirements, in the end, make sure that there are enough physical resources available, such as the required computing devices and a reliable internet connection for the uninterrupted development and ultimately, testing.

Figure 1: The process is based on the Agile Methodology, which is characterized by different iterations throughout its stages. The planning phase consists of taking the requirements from the Input phase and determining the features and user stories for the current iteration (or sprint). The design phase converts these requirements into technical specifications, which also includes making the entity-relationship diagrams for the MySQL database and crafting the interface (UI) and experience (UX) to guarantee that the system is user-friendly for both students and teachers. The development phase is concerned with coding, and within this phase, the team develops the software with the selected technologies. Testing Review and Feedback is a cyclical and crucial procedure: the developers and the stakeholders verify the system's functionality (for instance, checking that the automated progress tracking formula is accurate) and give their input to sharpen features, which frequently results

in returning to the design or development phases. The last stage is Deployment, which encompasses putting the Interactive E-LMS's final and stable version on the servers so that it can be used by the students and teachers of Paliparan III Senior High School.

Figure 1: The Output of this study is the complete, and fully functional system of Interactive E-Learning Management System with Automated Progress Tracking for CSS Students that directly results from the successful execution of the Process phase. This is the main deliverable that brings together all the essential knowledge and technical requirements stipulated. The system is interactive providing lively digital modules, quizzes, and assessments that are in line with the CSS curriculum. The most important feature is the Automated Progress Tracking, which assesses, logs, and gives feedback on the students' performance instantly, thus doing away with the manual record-keeping process. The system is implemented and functional, it has become the new platform for the transfer of knowledge and monitoring of performance for the intended students.

Figure 1: The evaluation of Ilearn will be guided by the ISO/IEC 25010 Software Quality Standard to ensure the system meets essential criteria for functionality and usability. The system will undergo a series of assessments after its development and initial deployment. It will be evaluated according to the various ISO 25010 quality characteristics including Functional Suitability, where the existence of different features, such as automated progress tracking, and interactive module, will be verified. Performance Efficiency will determine how fast and to what extent the system processes data and produces outputs, particularly when multiple branches use the system simultaneously. The Usability will be assessed through surveys and feedback from students and teachers regarding interface comprehension, navigability, and overall user experience. The system's Reliability will be observed in realistic scenarios where its stability will be evaluated over a certain period and also monitoring error-free or downtime data input consistency. Security is another important ISO parameter

that will evaluate the safeguarding of sensitive data related to students and tutors with an emphasis on data confidentiality, access rights, and protection of records at the branch level. The system's performance in terms of Maintainability and Portability will be tested so as to certify that it can be updated easily, debugged, and customized for ilearn system. This systematic assessment, utilizing the ISO 25010 framework, will enable the conclusive validation of ilearn effectiveness, quality, and conformity to the operational needs.

Expected Outcomes/Solutions

The primary output of this study is the development of Ilearn**:** Development of an Interactive E-Learning Management System with Automated Progress Tracking. For CSS Students of Paliparan III Senior Highschool Students. This system aims to overcome the difficulties encountered by both CSS students and teachers at Paliparan III Senior High School, thus facilitating a more efficient, well-organized, and data-based educational atmosphere. It is anticipated that the successful application of the finalized modules will yield specific outcomes and remedies as follows:

1. Secure Platform Administration and Organized Academic Structure

The system is capable of providing the basic solution for safe operation and orderly academic structure. The User Account Management module, which is an administrative tool, is responsible for creating users, assigning roles, and securing login for everyone. The Subject & Class Management module comes to help with this as it gives the Admin the possibility to establish all CSS subjects and appoint teachers thus forming a necessary structured academic system. The System Monitoring module which monitors total activity is responsible for the central organization being preserved and managed.

2.Core Content Delivery and Assessment Management

The result gives a thorough answer for the substitution of manual distribution and grading with interactive digital tools. The Teacher Modules grant teachers the privilege of managing their learning materials, permitting them to upload and sort all CSS lessons and modules (PDFs, videos) for student accessibility. The Activity & Assessment Management module provides the facility for the teachers to prepare quizzes and assignments, assign due dates, and work with either manual or system auto-grading functions, which will secure an uninterrupted evaluation process. The students interact with the curriculum through Learning Materials Access and Activity & Assessment Submission.

3. Interactive Learning Module

As a core solution designed to improve student engagement and understanding of CSS concepts, the system implements the interactive learmning module (Student) that provides the system's dedicated interactive learning component. These subject-based mini-activities reinforce lessons and help students review course content in an engaging manner, separate from formal, graded assignments.

4.Automated Progress Tracking and Performance Insight.

This module is the backbone of the solution that automates the monitoring of learner's progress, thus getting rid of manual recordings and allowing for timely intervention. The Student Automated Progress Tracking module records performance metrics in real time, producing a subject-wise visual progress bar that makes it possible for the students to see their activities done, quizzes taken with scores, and their general performance, thus promoting accountability. The Teacher Student Progress Monitoring module provides teachers with an up-to-the-minute and neatly arranged picture of every pupil's module completions, quiz scores, and

participation levels, which opens the door for speedy feedback and intervention actions.

5.Centralized User Dashboards and Communication Streamlining

The project is set up to ensure that communication is completely secure and centralized. One of the benefits of the Teacher Dashboard is that it quickly gives teachers an overview of what is going on in their classes, such as assignments, new submissions and student progress summaries. Meanwhile, the Student Dashboard displays all the subjects the student is currently enrolled in, the coming tasks and a quick view of the academic standing. Communication is done through Posting Announcements (Teacher) and Viewing Announcements (Student) in a very effective manner so that both teachers and students are notified on time and are informed about the class and provided with essential reminders. In the end, the system gets support from Admin Management Tools for total control over the maintenance tasks.

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